FRANCES (SU) LIU

Tel: +61 477 105 413 *Email*: Frances.Liu@uts.edu.au *Web*: frances-su-liu.github.io

RESEARCH INTERESTS

Financial econometrics; volatility modeling and forecasting; financial machine learning; high-frequency and alternative data; portfolio construction and risk modeling.

My research studies when richer information sets and more flexible models improve financial forecasts, and how those gains vary across market states. I am extending this work toward high-dimensional portfolio allocation and empirical asset pricing, where the same question governs whether added model complexity earns its place.

EDUCATION

Doctor of Philosophy (PhD) in Finance *2024.02–2027.08 (expected)*
University of Technology Sydney (UTS), Sydney, Australia

Supervisors: Assoc. Prof. Vitali Alexeev (UTS, Finance) & Assoc. Prof. Katja Ignatieva (UNSW, Actuarial)

Master of Science in Financial Engineering *2015.08–2017.06*
Stevens Institute of Technology, New Jersey, United States *GPA: 3.9/4.0*

Thesis: Systemic Risk Measurement: CoVaR and Copula Models (ARMA/TGARCH conditional moments, copula-based dependence across financial sectors, time-series CoVaR). Supervised by Assoc. Prof. Zhenyu Cui

Bachelor of Economics in Financial Engineering *2011.09–2015.06*
Harbin University of Commerce, Harbin, China

RESEARCH PAPERS

Reading Between the Trades: Neural Encoding Extended-Hours Tick Sequences for Volatility Forecasting
with Vitali Alexeev (UTS), Adam Clements (QUT), and Katja Ignatieva (UNSW) *working paper*

- Replaces the coarse overnight aggregates used in the volatility literature with the complete tick-level record of each extended-hours session, encoding variable-length, irregularly spaced transaction sequences and integrating the learned representations with a HAR benchmark
- Develops an incremental architecture hierarchy separating the contributions of non-linearity, tick-level information, trade weighting, and sequence ordering, which distinguishes information gains from gains attributable to model capacity
- Applied to all DJIA constituents from 2002 to 2025, reduces one-day-ahead out-of-sample forecast loss by approximately 11%, roughly twice the gain of the strongest scalar overnight proxy, with improvements stable across the Global Financial Crisis and COVID-19
- Finds that pre-open and post-close sessions generate comparable aggregate gains through different mechanisms: 64% of the pre-open improvement is recoverable from a within-session aggregate, while 81% of the post-close improvement is attributable to tick-sequence structure
- *Accepted*: Econometric Society Australasia Meeting (Adelaide, November 2026)
- *JEL*: C45, C53, G12, G14

When Does Sentiment Predict Volatility?
with Vitali Alexeev (UTS) and Katja Ignatieva (UNSW) *under review at the Journal of Applied Econometrics*

- Evaluates whether vendor-derived Thomson Reuters MarketPsych sentiment indices add predictive information to HAR volatility forecasts, and shows that modest average gains conceal substantial state dependence
- Finds that sentiment becomes more informative in the upper tail of the volatility distribution and when volatility persistence weakens, with persistence measured in a fractional-integration framework
- Shows that out-of-sample gains concentrate in jointly high-volatility and low-persistence regimes, consistent with sentiment supplying forward-looking narrative information when price-based predictors are least reliable
- *Presented*: International Association for Applied Econometrics (IAAE) Annual Conference (Carcavelos, June

2026)

- *Accepted*: Econometric Society Australasia Meeting (Adelaide, November 2026); FIRN Annual Conference (K'gari, November 2026)
- *JEL*: C22, C53, G12, G14

AWARDS AND SCHOLARSHIPS

UTS Business Doctoral Scholarship (full stipend)	2024–2027
UTS International Research Scholarship (full tuition)	2024–2027
Academic Excellence Award, Stevens Institute of Technology	2017

TEACHING EXPERIENCE

University of Technology Sydney — *Sydney, Australia*
Tutor, Finance Discipline Group — 2024–present

- 25503 *Investment Analysis* (undergraduate), Spring 2026
- 25556 *The Financial System* (undergraduate), Autumn and Spring 2026

INDUSTRY EXPERIENCE

Six years of applied quantitative research across systematic strategies, portfolio construction and asset allocation, risk modeling, structured-finance valuation, and research infrastructure, carrying models from hypothesis through backtesting to deployment.

Webull Financial LLC — *Changsha, China*
Senior Quantitative Analyst — 2020.04–2023.11

- Developed portfolio construction and asset allocation for a client-facing investment-advisory product spanning five risk profiles, combining screened ETF universes, mean-variance optimization, CAPM-implied returns, machine learning denoising, and covariance shrinkage
- Researched and implemented mid-frequency strategies across Chinese and US equity markets, including factor-based stock selection, sector allocation, ETF rotation, and equity-index strategies
- Took strategies from research and backtesting through paper trading to live deployment, with ongoing post-deployment performance monitoring, using S&P, FactSet, and Refinitiv data
- Designed and co-developed the shared research database and backtesting environment used by the 12-person private-fund quantitative research team

Pricing and Analytics (Shenzhen) Limited — *Shenzhen, China*
Quantitative Analyst: Derivatives Pricing and Valuation — 2017.12–2019.10

- Developed dynamic pricing, valuation, and risk models for property-backed REITs and CLOs issued by listed real-estate companies across successive issues; two models were registered as software copyrights
- Modeled underlying cash flows and calibrated prepayment assumptions from market-observable spreads, and applied Black-Scholes, risk-neutral and martingale pricing, extreme value theory, and Value at Risk for portfolio tail risk
- Implemented the pricing engines in C++ with Excel interfaces and VBA integration, with relative valuation against market comparables selected at city and listed-issuer level

COMPUTING AND LANGUAGES

Computing: Python (PyTorch, pandas, NumPy), GPU/HPC cluster computing, SQL, R, C++
Languages: Mandarin (native), Cantonese (fluent), English (fluent)